Atiyah-Singer Index Theorem for Dirac Operator

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The most prestigious theorem in differential topology would be the Index theorem, originally shown by Atiyah and Singer sixty years ago [1] . A mathematical approach to proving this requires the knowledge of exact sequences and a rich knowledge of de Rham cohomology and characteristic classes. Here, apart from being general, we are going to prove the Index theorem for Dirac operator in 4-dimensional spacetime. The generalization to higher dimensions follows similar steps.

I. INTRODUCTION

The Index theorem states that for the Dirac operator, the analytical index equals the topological index. In order to prove this theorem, we first have to bring up the notion of those two indices. The analytical index is closely related to the zeros (or the kernel) of the operator, while the topological index is a geometric invariant obtained by integrating some local details on a manifold. Such an idea was first proposed in Gauss-Bonnet theorem for 2-dimensional manifold as[2]

$$\int \epsilon^{ij} F_{ij} d^2x = 4\pi(1-g) \tag{1}$$

where F_{ij} is the second fundamental form and g is the genus of the manifold. In this article, our dimension will be 4, representing the spacetime.

To have a feeling of the index theorem, let us look at the degree of a continuous function $f : \mathbb{R} \to \mathbb{R}$. Let α be any bump 1-form then the degree is given by[3]

$$\int f^*\alpha. \tag{2}$$

If the degree is $1 \in \mathbb{Z}_2$, the function connects $-$ to $+$; If the degree is -1 , the function connectes $+$ to $-$ and otherwise the degree is 0. Meanwhile, if we count the number of ascending zeros n_+ and descending zeros n_-, we will have

$$\deg f = \int f^*\alpha = n_+ - n_-. \tag{3}$$

The degree is a topological index, because if we fix the two ends of the function and deform it, the index remains. The subtration of different types of zeros is the analytical index. $f(x) = x$ connects $-$ to $+$ and has one ascending zero so it has degree 1. $f(x) = x^2 - 1$ connects $+$ to $+$ and has one ascending, one descending zeros so the degree is 0.

Our full Index theorem is more complicated. We will first examine the mathematics of winding number, for it is the most fundamental object in topology. Then we will examine the chiral anomaly in gauge theories and see how these indices appear. By comparing and computing the indices, we will relate them to the analytical index and the topological index, and finally conclude that they are equal. As such, we prove the Index theorem for the Dirac operator.

II. WINDING NUMBER AND TOPOLOGY

In this section, we will examine the notion of winding number, one of the most important concept in topology. I am going to start with the simplest case: winding number in $\mathbb{C}$. Given a curve $\gamma : [0, 2\pi] \to \mathbb{C}$, we can count the number of times it winds around the origin. The can be done by the residue theorem:

$$\int_\gamma \frac{dz}{z} = 2\pi i \mathrm{Wind}[\gamma, 0] \tag{4}$$

so

$$\mathrm{Wind}[\gamma, 0] = \frac{1}{2\pi i}\int_\gamma \frac{dz}{z} = \frac{1}{2\pi i}\int_\gamma d\log z. \tag{5}$$

In the following context, I will only talk about the winding number around the origin, because otherwise we only need to do a change of variables to shift back to the origin, and the winding number will be denoted by $W(\gamma)$ or W or even n. Where is the topology in this? The homotopy theorem tells us that the integral remains unchanged if γ undergoes a homotopic transformation. In fact, this has a deeper application in condensed matter physics, known as the topological insulators.

A vector bundle is twisted (or non-trivial) if it does not admit a global trivialization. This is to say it has to break in to several patches and at the patching boundary the transition map must not be homotopic to identity. If the patching boundary is a line C, that is, 1-dimensioanl, we have the first Chern number

$$\frac{1}{2\pi}\int c_1 = \frac{1}{2\pi i}\int_C d\log\det U = W \tag{6}$$

where U is the transition map. This is exactly the winding number of the phase of the determinant of U. This is a long sentence, but is easy to grasp. If U is homotopic to identity then its phase along C should be homotopic to 1, which does not wind around the origin. Otherwise, we have a non-trivial bundle. In condensed matter theory the base space is the Brillouin zone $\mathbb{T}^2$ and the fiber being $U(1)$. Since $c_1 = e$, the Euler class so

$$\int c_1 = 2\pi\chi \tag{7}$$

where χ is the Euler characteristic of the bundle. By Poincar-Hopf theroem, the Euler characteristic is the

sum of the winding number of the function around a small sphere (the index, or degree of a map) centered at the zeros. Thus the first Chern number can be written in terms of integrals or equivalently the total index. This is a first glance at the Index Theorem.

In order to push further, we need the notion of a generalized winding number. Since we know as a famous result that $\pi_n(S^n) = \mathbb{Z}$, for each dimension we can define a winding number. A slick way to define the winding number is that, the oriented surface area of an n-fold unit sphere is n times the surface area of a unit sphere. Thus we normalized each closed manifold and compute its surface area. Let us look at the example for 1-dimensional hypersurface in 2 dimensions—just a curve:

$$L = \int_\gamma \hat{z} \times \mathbf{r}' \cdot \hat{\mathbf{r}} dt = \int_\gamma \epsilon^{ij} \frac{\mathbf{r}_i}{r} d\mathbf{r}_j \tag{8}$$

Now, if we normalize γ which amounts to mapping all $\mathbf{r}$ to $\mathbf{r}/\hat{\mathbf{r}}$, the oriented length becomes

$$L = \int_\gamma \frac{\epsilon^{ij} \mathbf{r}_i d\mathbf{r}_j}{r^2} \tag{9}$$

where we used the fact that

$$d\left(\frac{\mathbf{r}_i}{r}\right) = \frac{r d\mathbf{r}_i - \mathbf{r}_i dr}{r^2} \tag{10}$$

with the second term

$$\int \epsilon^{ij} \frac{\mathbf{r}_i \mathbf{r}_j}{r^2} dr = 0. \tag{11}$$

Since the oriented length of $\mathbb{S}^1$ is 2π, the winding number of a curve is then

$$\frac{1}{2\pi} \int_\gamma \frac{\epsilon^{ij} \mathbf{r}_i d\mathbf{r}_j}{r^2} = \frac{1}{2\pi} \int_\gamma \frac{x dy - y dx}{r^2}. \tag{12}$$

One immediately realizes that this is the real version of the complex winding number with $z \mapsto x + iy$.

Now, for $M^2 \subset \mathbb{R}^3$, we have the winding number to be

$$\frac{1}{4\pi} \int \hat{\mathbf{r}} \cdot (\partial_u \hat{\mathbf{r}} \times \partial_v \hat{\mathbf{r}}) = \frac{1}{8\pi} \int_M \frac{\epsilon^{ijk} \mathbf{r}_i d\mathbf{r}_j \wedge d\mathbf{r}_j}{r^3}. \tag{13}$$

where 4π is the oriented surface area of $\mathbb{S}^2$ and the factor $1/2 = 1/(2!)$ is obtained when written with the differential form.

By inducting on the dimension, we have that the winding number for $M^n \subset \mathbb{R}^{n+1}$ is

$$W = \frac{1}{n! |\mathbb{S}^n|} \int_M \frac{\epsilon^{ij\cdots k} \mathbf{r}_i d\mathbf{r}_j \wedge \cdots \wedge d\mathbf{r}_k}{r^n}. \tag{14}$$

One can also obtain the integrand form the n-volume form embedded in $\mathbb{R}^{n+1}$:

$$d\omega_{\text{Vol}} = \sum_j (-1)^{j-1} x_j dx^1 \wedge \cdots \wedge \cancel{dx^j} \wedge dx^n. \tag{15}$$

III. CHIRAL ANOMALY

In this section, we will derive the index for Dirac operator $i\not{D}$ and also in an integral form, just like the relation in Poincar-Hopf theorem. Eventually, this leads to the index theorem for the Dirac operator. Before that, we have to review on some Grassmannian functional integral.

A. Grassmannian Integral

We define anti-commuting objects $\theta_1, ..., \theta_n$ over the complex field[4]. They form a vector field. Let the space of all polynomials in them be Λ^n. Then we may define integrals over Λ^n, the functional space to be

$$\int_{\Lambda_n} \theta_1 \theta_2 \cdots \theta_n d\theta = 1, \quad \int_{\Lambda_n} \text{not enough} d\theta = 0. \tag{16}$$

Now comes the change of variable integral.

$$\int_{\Lambda_n} f \circ U(\theta_1, ..., \theta_n) d\theta \tag{17}$$

where $U : \theta_i \mapsto U(\theta_i) := u_i$. Then we have

$$\int_{\Lambda_n} f(u_1, ..., u_n) |\det U| du = \int_{\Lambda_n} f \circ U(\theta_1, ..., \theta_n) d\theta \tag{18}$$

The proof goes as: Let

$$f = C u_1 \cdots u_n, \tag{19}$$

then if $u_i = U_{ij} \theta_j$ then

$$f = C U_{1i} u_i \cdots U_{nj} u_j = C \det(U) \theta_1 \cdots \theta_n. \tag{20}$$

Now,

$$\int_{\Lambda_n} f \circ U(\theta_1, ..., \theta_n) d\theta = C \det(U) \tag{21}$$

and

$$\int_{\Lambda_n} f(u_1, ..., u_n) |\det U| du = C \det(U). \tag{22}$$

Thus

$$\int_{\Lambda_n} f(u_1, ..., u_n) du = \int_{\Lambda_n} \tilde{f}(\theta_1, ..., \theta_n) |\det U(\theta)|^{-1} d\theta \tag{23}$$

B. Functional Integral

We define

$$\int D\psi = \int \prod_x d\psi_x \tag{24}$$

Now, how do we deal with change of variable of ψ? Let

$$\psi(a) = a_n \phi_n, \tag{25}$$

then

$$\psi_x = \phi_{xn} a_n \tag{26}$$

$$\int d\psi = \int (\det \phi_{xn})^{-1} da. \tag{27}$$

Here comes something fishy. x is continuous and takes uncountably many values while n seems to be discrete and takes countably many values. This is actually not ture. For example the wave equation $(\partial_x^2 - \partial_t^2)\psi = 0$ has uncountably many eigenfunctions. To resolve, let us look at the operator D or $\not{D}$ and discretize our space: Let functions only take value on N numbers of x-points. Then each function $\psi(x)$ is in fact, a column vector

$$[\psi(x_1), ..., \psi(x_N)]\,. \tag{28}$$

Then our operator D is $V \to V$ where V is N-dimensional. And of course, just like our process of diagonalization $A = Q\Delta Q^\dagger$, the matrix ϕ_{xn} plays the same role as Q. As expected, this can always be taken determinant with norm 1.

C. Integral From of Dirac Operator

We are goin to examine the chiral symmetry:

$$\begin{aligned} \psi \mapsto \psi' &= e^{i\alpha\gamma_5}\psi = \psi + i\alpha\gamma_5\psi, \\ \bar\psi = \psi^\dagger\gamma^0 \mapsto \bar\psi' &= \bar\psi e^{i\alpha\gamma_5} = \bar\psi + \bar\psi i\alpha\gamma_5. \end{aligned} \tag{29}$$

Let the initial wavefunctions be written in the eigenbasis of $\not{D}$:

$$\psi = \sum_n a_n \phi_n(x), \quad \bar\psi = \sum_n \bar b_n \phi_n^\dagger(x) \tag{30}$$

then

$$\begin{aligned} a_n' &= \int \phi_n^\dagger(x)\psi'(x) d^4x \\ &= a_n + \sum_m \int \phi_n^\dagger(x) i\alpha\gamma_5 a_m \phi_m(x) d^4x \end{aligned} \tag{31}$$

and

$$\begin{aligned} \bar b_n' &= \int \bar\psi'(x)\phi(x) d^4x \\ &= \bar b_n + \sum_m \int \bar b_m \phi_m^\dagger(x) i\alpha\gamma_5 \phi_n(x) d^4x \end{aligned} \tag{32}$$

The Lagrangian respects chiral symmetry so the action is unchanged. However, our measure changes:

$$D\bar\psi' D\psi' = \int \prod_i d\bar b_i da_i \det\left(\delta_{mn} + \int \phi_m^\dagger(x) i\alpha\gamma_5 \phi_n(x) d^4x\right)^{-2} \tag{33}$$

Now we will have to use the identity

$$\det(I + \alpha A)^{-2} = \det e^{-2\log(I+\alpha A)} = \det e^{-2\alpha A} = e^{-2\alpha \mathrm{Tr} A}. \tag{34}$$

and evaluate the trace of $-2i\alpha\gamma_5$. The trace does not depend on the choice of basis, so we take the Fourier basis for convenience.

$$\mathrm{Tr}\gamma_5 = \mathrm{Tr}\int \frac{d^4k}{(2\pi)^4} \int d^4x e^{-ikx}\gamma_5 e^{ikx} \tag{35}$$

Because we do not know if this expression converges, we have to adopte a smooth regulator $f(x) : f(0) = 1$ and $f(\infty) = 0$ with $f'(x) = 0$ on $[0, 1]$ [5]. An example of such a function is

$$f(x) = \begin{cases} 1 & x \in [0,1] \\ 1 - e^{-\frac{1}{(x-1)^2}} & x \in [1, \infty) \end{cases} \tag{36}$$

whose graph looks like

Now the trace becomes

$$\lim_{M\to\infty} \mathrm{Tr}\int d^4x \int \frac{d^4k}{(2\pi)^4} e^{-ikx}\gamma_5 f(\not{D}^2/M^2) e^{ikx} \tag{37}$$

with

$$\not{D}^2 = D_\mu D^\mu - \frac{i}{4}[\gamma^\mu, \gamma^\nu] F_{\mu\nu}. \tag{38}$$

Thus

$$\begin{aligned} &\int d^4x \mathrm{Tr}\int \frac{d^4k}{(2\pi)^4} e^{-ikx}\gamma_5 f(\not{D}^2/M^2) e^{ikx} \\ &= \int d^4x \mathrm{Tr}\int \frac{d^4k}{(2\pi)^4}\gamma_5 f\left(-\frac{k_\mu k^\mu}{M^2} - \frac{i}{4M^2}[\gamma^\mu, \gamma^\nu] F_{\mu\nu}\right). \end{aligned} \tag{39}$$

M will eventually be infinite so we expand f in powers of $1/M^2$ [5]:

$$\begin{aligned} &f(-\frac{k^2}{M^2} - \frac{C}{M^2}) \\ &= f(-\frac{k^2}{M^2}) - f'(-\frac{k^2}{M^2})\frac{C}{M^2} + \frac{1}{2!}f''(-\frac{k^2}{M^2})\frac{C^2}{M^4}. \end{aligned} \tag{40}$$

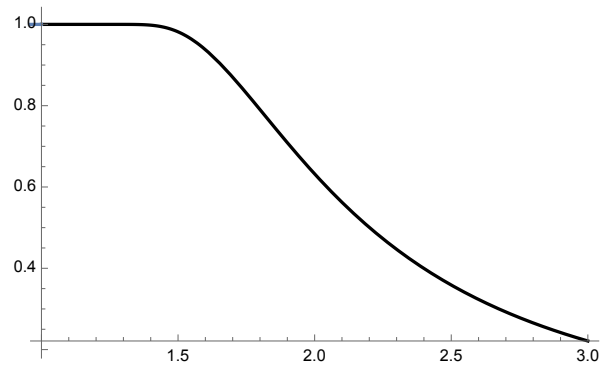

FIG. 1. Graph of the C^∞ regulator function, whose value is a constant 1 on [0,1].

The first terms vanishes because $\mathrm{Tr}\gamma_5 = 0$, and the second term vanishes because f' is strictly vanishing near 0. What it left is:

$$\mathrm{Tr}\int \frac{d^4k}{(2\pi)^4}\gamma_5 f''\left(-k^2\right)\left(\frac{i}{4}[\gamma^\mu,\gamma^\nu]F_{\mu\nu}\right)^2\frac{1}{M^4}, \tag{41}$$

Now, we do a change of variable: $k \mapsto Mk$ then the integral becomes

$$\begin{aligned}&\mathrm{Tr}\int \frac{d^4k}{(2\pi)^4}M^4\gamma_5 f''\left(-k^2\right)\left(\frac{1}{4}[\gamma^\mu,\gamma^\nu]F_{\mu\nu}\right)^2\frac{1}{2M^4}\\ &\quad=\frac{1}{32}\mathrm{Tr}\int \frac{d^4k}{(2\pi)^4}\gamma_5 f''(-k^2)\left([\gamma^\mu,\gamma^\nu]F_{\mu\nu}\right)^2.\end{aligned} \tag{42}$$

From this we can see that higher orders must vanish because they have higher orders of $1/M$. Now, we have to compute

$$\begin{aligned}&\mathrm{Tr}\gamma_5\left([\gamma^\mu,\gamma^\nu]F_{\mu\nu}\right)^2\\ =&F_{\mu\nu}F_{\alpha\beta}\mathrm{Tr}\gamma_5(\gamma^\mu\gamma^\nu-\gamma^\nu\gamma^\mu)(\gamma^\alpha\gamma^\beta-\gamma^\beta\gamma^\alpha)\\ =&4F_{\mu\nu}F_{\alpha\beta}(\epsilon^{\mu\nu\alpha\beta}-\epsilon^{\nu\mu\alpha\beta}-\epsilon^{\mu\nu\beta\alpha}+\epsilon^{\nu\mu\beta\alpha})\\ =&16F_{\mu\nu}F_{\alpha\beta}\epsilon^{\mu\nu\alpha\beta}=32F_{\mu\nu}\tilde{F}^{\mu\nu}\end{aligned} \tag{43}$$

where $\tilde{F}^{\mu\nu}=\frac{1}{2}\epsilon^{\mu\nu\alpha\beta}F_{\alpha\beta}$ is the Hodge dual of F. The integral now becomes

$$\int d^4x\tilde{F}F\int\frac{d^4k}{(2\pi)^4}f''(-k^2) \tag{44}$$

where the latter integral is

$$\int\frac{d^4k}{(2\pi)^4}f''(-k^2)=\frac{1}{16\pi^2}\int_0^\infty k^3 f''(-k^2)dk. \tag{45}$$

Let $u=-k^2$ and integrate by parts, we obtain

$$\begin{aligned}&\int_{-\infty}^0 -uf''(u)du=\int_0^\infty uf''(u)du\\ &=uf'(u)du\Big|_0^\infty-\int_0^\infty f'(u)du=1.\end{aligned} \tag{46}$$

Combining everything, we eventually can say that,

$$\mathrm{Tr}\gamma_5=\int d^4x\frac{1}{16\pi^2}\tilde{F}F, \tag{47}$$

and the chiral anomaly will multiply an extra term[4]

$$e^{iS}\mapsto e^{iS}\exp(-2i\alpha\mathrm{Tr}\gamma_5). \tag{48}$$

IV. INDEX THEOREM

It turns out that, $\mathrm{Tr}\gamma_5$ is an integer. But in fact, there are two different ways to see this, and this will be intepreted as the Index theorem eventually. The Index theorem states that, the topological index equals the analytical index.

In the following sections, we will show that

$$\int d^4x\frac{1}{16\pi^2}\tilde{F}F \tag{49}$$

is the topological index, and

$$\mathrm{Tr}\gamma_5 \tag{50}$$

is the analytical index. They are both integers. As shown in the previous section,

$$\mathrm{Tr}\gamma_5=\int d^4x\frac{1}{16\pi^2}\tilde{F}F \tag{51}$$

so we will show that the analytical index equals the topological index.

A. Topological Index

The Chern-Simons form Ω is defined such that

$$d\Omega=F^2d\omega_{\mathrm{Vol}}=\tilde{F}Fd^4x \tag{52}$$

from the definition of Hodge dual. Thus

$$\mathrm{Tr}\gamma_5=\frac{1}{16\pi^2}\int d\Omega. \tag{53}$$

Now things become much more interesting. The Chern-Simons form is a characteristic form, and produces a geometric invariant via integral over the tangent bundle fiber[6]. Here

$$\Omega=G_\mu\wedge dx^\mu,\quad G_\mu=\epsilon^{\mu\nu\lambda\sigma}\mathrm{Tr}A_\nu\wedge\left(F_{\lambda\sigma}-\frac{2}{3}A_\lambda A_\sigma\right). \tag{54}$$

By Stokes theorem,

$$\int_M d\Omega=\int_{\partial M}\Omega. \tag{55}$$

Here ∂M is a 3-dimensional manifold. In vaccum, we have A as a pure gauge at the boundary so

$$A=UdU^{-1},\quad F=0. \tag{56}$$

Suppose $U \in G = SU(2)$, we can write $U = r_0 + ir_j\sigma^j$. The integral becomes

$$\begin{aligned} &-\frac{2}{3}\int_{\partial M} \epsilon^{\mu\nu\lambda\sigma}\hat{r}_\mu \mathrm{Tr} A_\nu \wedge A_\lambda \wedge A_\sigma \\ &= \frac{4}{3}\int_{\partial M} \epsilon^{\mu\nu\lambda\sigma}\hat{r}_\mu \frac{dr_\nu \wedge dr_\lambda \wedge dr_\sigma}{r^3} \end{aligned} \tag{57}$$

where we used that $\sigma_1\sigma_2\sigma_3 = i$. One immediately realize that this is the expression for the winding number! Thus we integrate out to obtain

$$\frac{4}{3} \times 3! \times |\mathbb{S}^3| \times n = 16\pi^2 \tag{58}$$

and so

$$\frac{1}{16\pi^2}\int d\Omega = n, \tag{59}$$

the winding number and is a topological quantity. These arguments can let us find the identity map: We just need (r_0, r_1, r_2, r_3) to wind around the origin for *one* time. So just let them be $r_i = x_i/r$, leading to

$$U = \frac{x_0 + \vec{x}\cdot\vec{\sigma}}{r} \tag{60}$$

and is known as the instanton [7]. A larger winding number can be easily constructed by mapping it many times as U^n.

The significance of all above is that $\tilde{F}F$ is an exact form $d\Omega$. And for

$$\int_M d\Omega, \tag{61}$$

if a homotopy H acts on Ω by a pullback[3],

$$\int_M dH^*\Omega = \int_M H^* d\Omega = \int_{H(M)} d\Omega. \tag{62}$$

So the effect of the homotopy is to deform the space M. However, we just shown that (at least for $SU(2)$), that the integral is a topological quantity so will not change if M undergoes a continuous deformation. Therefore we are just to classify the homotopy classes.

The space we are integrating over is $\mathbb{S}^3$ and the integrand is a function of U in some Lie group G. Thus we are looking for the third homotopy class of G—$\pi_3(G)$. When $G = SU(2) \simeq \mathbb{S}^3$, $\pi_3(G) = \pi_3(\mathbb{S}^3) = \mathbb{Z}$, consisting of the winding numbers. $SU(2)$ is just the electroweak theory. For QCD, $G = SU(3)$ and $\pi_3(SU(3)) = \mathbb{Z}$, too[8]. We would definitely expect $\pi_3(SU(3)) \supseteq \pi_3(SU(2)) = \mathbb{Z}$ and luckily they are equal. Furthermore $\pi_3(SU(4)) = \mathbb{Z}$. For Standard Model, we have

$$\pi_3(SU(3) \times SU(2) \times U(1)) = \mathbb{Z} \oplus \mathbb{Z}. \tag{63}$$

B. Analytical Index

The analytical index, on the other hand, is more direct to compute. It is defined for any operator O as

$$\mathrm{dimKer}(O) - \mathrm{dimKer}(O^\dagger). \tag{64}$$

Let us recall

$$\mathrm{Tr}\gamma_5 = \sum_n \int d^4x \phi_n^\dagger(x)\gamma_5\phi_n(x). \tag{65}$$

Let $\not{D}\phi = E\phi$ then

$$\not{D}\gamma_5\phi = -\gamma_5\not{D}\phi = -E\gamma_5\phi. \tag{66}$$

Therefore, $\gamma_5\phi$ is also an eigenstate, but with an opposite energy. This would mean that ϕ and $\gamma_5\phi$ are orthogonal so

$$\int d^4x \phi_n^\dagger(x)\gamma_5\phi_n(x) = 0. \tag{67}$$

Summing over n, the contributing terms must be those ϕ_n with 0 energy, i.e. $\not{D}\phi = 0$. If $\gamma_5\phi = \phi$,

$$\int d^4x \phi_n^\dagger(x)\gamma_5\phi_n(x) = 1 \tag{68}$$

and if $\gamma_5\phi = -\phi$,

$$\int d^4x \phi_n^\dagger(x)\gamma_5\phi_n(x) = -1. \tag{69}$$

Therefore summing those up yields

$$n_+ - n_- \tag{70}$$

where n_+ is the number of left-handed particles and n_- right-handed with vanishing energy. Thus let the operator

$$O = \frac{1-\gamma_5}{2}\not{D} = \not{D}_+, \quad O^\dagger = \frac{1+\gamma_5}{2}\not{D} = \not{D}_-, \tag{71}$$

the analytical index becomes

$$\begin{aligned} &\mathrm{dimKer}(O) - \mathrm{dimKer}(O^\dagger) \\ =&\mathrm{dimKer}(\not{D}_+) - \mathrm{dimKer}(\not{D}_-) \\ =&n_+ - n_-. \end{aligned} \tag{72}$$

V. CONCLUSION

In this article, we have shown the well-known Atiyah-Singer Index theorem for this particular Dirac operator. We started with the operator $\not{D} = \gamma^\mu(\partial_\mu - iA_\mu)$ and derived the equality between two different expressions $\mathrm{Tr}\gamma_5$ and

$$\frac{1}{16\pi^2}\int F^2, \tag{73}$$

which arises from the chiral anomaly of a gauge group G theory. This term is a topological quantity and is related to $\pi_3(G)$. Then we showed that $\mathrm{Tr}\gamma_5$ is another quantity that is intrinsic in the zeros of the operator, expressed by

$$\mathrm{dimKer}(\not{D}_+) - \mathrm{dimKer}(\not{D}_-). \tag{74}$$

Of course, one could work with $2n$ dimensional space and compute the Chern-Simons form $d\Omega$. This will give us the generalization in the dimensions.

[1] M. F. Atiyah (1997), URL https://api.semanticscholar.org/CorpusID:120233267.

[2] M. P. do Carmo, *Differential geometry of curves and surfaces.* (Prentice Hall, 1976), ISBN 978-0-13-212589-5.

[3] R. Bott and L. W. Tu, *Differential Forms in Algebraic Topology* (Springer, 1982), ISBN 978-0-387-90613-3, 978-1-4419-2815-3, 978-1-4757-3951-0.

[4] M. E. Peskin and D. V. Schroeder, *An Introduction to quantum field theory* (Addison-Wesley, Reading, USA, 1995), ISBN 978-0-201-50397-5.

[5] K. Fujikawa and H. Suzuki, *Path integrals and quantum anomalies* (2004).

[6] S.-S. Chern and J. Simons, Annals of Mathematics **99**, 48 (1974), ISSN 0003486X, URL http://www.jstor.org/stable/1971013.

[7] S. Coleman, *Aspects of Symmetry: Selected Erice Lectures* (Cambridge University Press, 1985).

[8] M. Mimura and H. Toda, Journal of Mathematics of Kyoto University **3**, 217 (1963), URL https://doi.org/10.1215/kjm/1250524818.